

Solution Requirements

Date	07 July 2026
Team ID	SWTID-2026-3218
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Admin/Analyst/Customer) can securely log in using username and password.	High
2	Authorization levels	Admin manages users and models, analysts review applications, customers can check eligibility.	High

3	External interfaces	Connects with Kaggle dataset, Flask web application, IBM Watson Machine Learning Cloud, and browser interface.	High
4	Transactions processing	Accept applicant details, preprocess data, run ML model, and generate approval/rejection prediction.	High
5	Reporting	Display prediction result, model accuracy, confusion matrix, classification report, and data visualizations.	Medium
6	Business rules, etc.	Validate user inputs, reject incomplete applications, preprocess data before prediction, and use the best-performing ML model.	High
7	Compliance to laws or regulations	Protect applicant data, maintain confidentiality, and follow basic data privacy and banking guidelines.	High
8	<i>Other</i>	Support real-time prediction, model comparison, and cloud deployment using IBM Watson ML.	Medium

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	Prediction results should be generated quickly.	Response time less than 2 seconds
2	Scalability	System should support multiple users and future dataset growth.	Support 1000+ users
3	Security & Data Privacy	Secure login, protect applicant information, and encrypt sensitive data.	Password encryption and secure data handling
4	Reliability & Availability	System should remain available with minimal downtime.	99% uptime
5	Usability & Accessibility	Easy-to-use web interface with simple forms and clear prediction results.	User-friendly interface with minimal navigation
6	<i>Other</i>	Maintainability and portability across Windows, Linux, and macOS.	Easy updates and cross-platform compatibility